COMPILING THE ALMGREN–PITTS COMBINATORIAL ARGUMENT

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Abstract. The Almgren–Pitts combinatorial argument is a recurring technical step in min-max constructions of embedded minimal hypersurfaces: a cover-refinement on the parameter interval, followed by a bookkeeping argument that turns local near-critical savings into a global supremum improvement. Each author re-derives it in local notation. We extract it from the literature as a standalone metric-combinatorial algorithm and formalize it in Lean 4 as the importable library `CombArg`.

1. Introduction

A minimal hypersurface is a critical point of the area functional in codimension one. On a closed Riemannian manifold (M^{n+1}, g) with $H_n(M;\mathbb{Z}) = 0$ (for instance $M = S^{n+1}$), direct minimization of the n-dimensional Hausdorff measure over integral currents yields only the empty hypersurface; a non-trivial minimal hypersurface must be constructed as a saddle-type critical point of the area functional. The construction, due to Almgren and Pitts, is a *min-max* scheme on varifolds; regularity of the output is due to Pitts [Pit81] and Schoen–Simon [SS81].

Theorem 1.1 (Almgren–Pitts min-max theorem [Pit81, SS81])**.** *Every closed Riemannian manifold (M^{n+1}, g) admits a nontrivial embedded minimal hypersurface $\Sigma \subset M$ whose singular set $\operatorname{Sing}\Sigma$ has Hausdorff dimension at most $n-7$.*

A key step in Pitts's proof, recurring in [CDL03, DLT13, DLR18, DL16, MN14], is the Almgren–Pitts combinatorial argument, stated in the Colding–De Lellis formulation as follows.

Theorem 1.2 (Almgren–Pitts combinatorial Argument, [CDL03, Prop. 3.4])**.** *Let Λ be a homotopically closed family of sweepouts of M^{n+1}, and $m_0 = \inf_\Lambda \mathcal{F}$. There exists a min-max sequence $\{\Gamma^N\}$ converging as varifolds to a stationary integer rectifiable varifold of mass m_0, with Γ^N $1/N$-almost-minimizing in every pair $(U^1, U^2) \in \mathcal{CO}$ of $\mathcal{H}^n$-disjoint open sets, for all N sufficiently large.*

The proof of Theorem 1.2 reduces, following [DLT13, §3.2], to the following claim on the sweepout:[1]

> **Claim.** For each N sufficiently large, there exists $t_N \in [0,1]$ such that $\Gamma^N := \Gamma^N_{t_N}$ is $1/N$-almost-minimizing in every $(U^1, U^2) \in \mathcal{CO}$ and $\mathcal{H}^n(\Gamma^N) \geq m_0 - 1/N$.

The Claim is proved by contradiction: from its negation, every t with $\mathcal{H}^n(\partial\Omega^N_t) \geq m_0 - 1/N$ admits a *local replacement* with saving $1/(4N)$ [DLT13, Lemma 3.1]; a combinatorial cover-refinement and bookkeeping argument on $[0,1]$ then assembles from these local replacements a modified sweepout with supremum strictly below m_0, contradicting $m_0 = \inf_\Lambda \mathcal{F}$. The combinatorial argument depends on no geometric measure theory, no admissible class, and no ambient regularity; it is a bookkeeping construction on continuous real-valued functions, yet has no canonical reference.

The combinatorial core of the proof of the Claim, which is the precise theorem we formalize, reads as follows.

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[1] Throughout this paper, "sweepout" denotes a one-parameter family $\{\Gamma_t\}_{t\in[0,1]}$ in the sense of [DLT13, Def. 3.1]; the multi-parameter min-max setting of [MN14] requires separate combinatorial machinery and lies outside our scope.

Theorem 1.3 (Combinatorial core)**.** *Let $f : [0,1] \to \mathbb{R}$ be continuous with $m_0 = \sup f$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose that at every $t \in [0,1]$ with $f(t) \geq m_0 - 1/N$ there is a* local witness*: an open neighborhood $U_t \ni t$ and a continuous replacement energy $E_t : [0,1] \to \mathbb{R}$ satisfying $f(s) - E_t(s) \geq 1/(4N)$ for every $s \in U_t$. Then there exist a finite family of closed pieces $\{p_l \subseteq [0,1]\}_{l\in\iota}$, continuous replacement energies $\{E_l : [0,1] \to \mathbb{R}\}_{l\in\iota}$, and savings $\{s_l \geq 1/(4N)\}_{l\in\iota}$ satisfying:*

(I) *$f - E_l \geq s_l$ on p_l;*
(II) *every $t \in [0,1]$ belongs to at most two of the pieces p_l;*
(III) *$\{t \in [0,1] : f(t) \geq m_0 - 1/N\} \subseteq \bigcup_l p_l$.*

Remark 1.4. A natural compactness sketch of Theorem 1.3 runs: $f^{-1}([m_0 - 1/N, \infty))$ is compact, so finitely many witness neighborhoods cover it; pass to an irredundant subfamily and take $p_l := \overline{U_{t_i}}$. Conditions (I) and (III) hold immediately; (II) does not. Even in dimension one, sub-selection alone need not deliver multiplicity ≤ 2 on *closed* pieces: $I_1 = (0, 0.4)$, $I_2 = (0.3, 0.5)$, $I_3 = (0.4, 0.7)$ together cover $[0.05, 0.65]$, every member is needed, and the open multiplicity is ≤ 2, yet $0.4 \in \overline{I_1} \cap \overline{I_2} \cap \overline{I_3}$. The witness hypothesis offers no leverage to avoid such configurations: nothing constrains the U_t to be spaced.

The non-trivial content of Theorem 1.3 is the *construction* of a cover $\{p_l\}$ which precludes them. A Lebesgue-number argument on the compact near-critical set produces an initial cover by intervals $\{I_i\}$ of controlled radius satisfying the skip-2 spacing condition (a) of [DLT13], $\overline{I_i} \cap \overline{I_{i+2}} = \emptyset$ — exactly ruling out the boundary collision above. A bounded smallest-index induction refines $\{I_i\}$ to pieces $\{J_k\}$, where each new piece is either I_{i^*} or $I_{i^*} \setminus I_{\sigma(\ell)}$; the set-difference branch is what makes this strictly more than sub-selection. A parity argument on (a) then transports the skip-2 disjoint-closure property to the two-fold overlap (II) on $\{\overline{J_k}\}$; continuity extends the per-piece saving from J_k to $\overline{J_k}$. The bookkeeping corollary below (the existence of a competitor f' with reduced supremum) is a three-line arithmetic consequence.

Remark 1.5 (Why the construction, not just the conclusion)**.** At the abstract scalar level alone, Theorem 1.3 admits a strictly shorter proof: a finite subcover by witness intervals, refined to a partition by collecting all interval endpoints, yields closed pieces of multiplicity ≤ 2 (sharing only endpoints) carrying the witness data. Such a partition suffices for the scalar conclusion because the multiplicity bound (II) is structurally recorded but *not arithmetically load-bearing* for $\sup f' \leq m_0 - 1/(4N)$ — any cover with per-piece saving $\geq 1/(4N)$ delivers (c) of Corollary 1.6 (`Cover.lean` states this explicitly).

What a partition discards is precisely what the formalization is meant to extract. The Almgren–Pitts combinatorial argument as it appears in [CDL03, DLT13, DLR18, DL16, MN14] is not the scalar conclusion of Theorem 1.3 but the *construction* that proves it: overlapping intervals $\{I_i\}$ with spacing condition (a), an injective refinement σ, and the parity rescue of two-fold overlap on closures. The geometric lift in the ambient GMT setting — DLT's modified sweepout $\{\partial\Omega'_t\}$, defined by blending two replacements on $J_i \cap J_{i+1}$ — requires *positive-measure* overlap and breaks at a partition's measure-zero endpoint sharing (the blend $\Omega'_t = [\Omega_t \setminus (U'_i \cup U'_{i+1})] \cup [\Omega_{i,t} \cap U'_i] \cup [\Omega_{i+1,t} \cap U'_{i+1}]$ needs both replacements simultaneously available on a region of positive t-measure to remain t-continuous).

The library therefore ships a two-tier architecture: the DLT-faithful path under `CombArg.OneDim` produces a structured `DLTCover` that exposes the spacing / refinement / σ-injectivity data a geometric consumer needs, with the abstract scalar theorem obtained as a downgrade; the partition shortcut is also formalized, as `CombArg.Scalar.exists_refinement_partition`, specifically as evidence that the DLT machinery is dispensable at the abstract scalar level alone — a dependency-graph fact verified by `lake build`: the partition module imports neither `OneDim/SpacedIntervals` nor `OneDim/Induction`.

Translation from the De Lellis–Tasnady setting to Theorem 1.3: the boundary $\partial\Omega_t$ becomes a continuous $f : [0,1] \to \mathbb{R}$; the Hausdorff-measure hypothesis becomes $m_0 = \sup f$; the nested pair (A_t, B_t, U_t') collapses to a single neighborhood U_t; the replacement family $\{\partial\Omega_{t,\tau}\}$ collapses to a single continuous E_t; and the almost-minimizing hypothesis, which in the original proof is the input to Lemma 3.1 of [DLT13], is absorbed into the existence of the local witness (the caller must supply witnesses wherever they would be produced in the ambient GMT setting). The output of Theorem 1.3 is the combinatorially assembled cover data $\{(p_l, E_l, s_l)\}$; the modified sweepout in the original proof is the geometric lift of the scalar competitor produced by the corollary below.

Corollary 1.6 (Sup reduction)**.** *Under the hypotheses of Theorem 1.3, with the pieces $\{p_l\}$ produced there, there exists $f' : [0,1] \to \mathbb{R}$ satisfying:*

(a) $f'(t) \le f(t)$ *for every* $t \in [0,1]$*;*
(b) $f'(t) = f(t)$ *whenever* $t \notin \bigcup_l p_l$*;*
(c) $\sup f' \le m_0 - 1/(4N)$.

Remark 1.7 (Why (a)+(b) are stated)**.** The bound (c) alone is trivialized by any constant $f' \equiv c \le m_0 - 1/(4N)$. Condition (a) ties f' pointwise below f; condition (b) forces f' to coincide with f wherever no piece modifies it. Together they anchor f' to f, so generic non-constant f rules out constant competitors, and (c) becomes a non-trivial quantitative assertion about the specific competitor the proof produces (concretely, $f'(t) = f(t) - \sum_{l:\, t\in p_l} s_l$).

The bookkeeping underlying Corollary 1.6 is strictly more general than the one-parameter statement it certifies: the same arithmetic applies to any compact, nonempty topological space K with continuous $f : K \to \mathbb{R}$ and any scalars (δ, ε) with $0 < \varepsilon \le \delta$, taking as input any finite cover of the δ-super-level set of f with per-piece savings $\ge \varepsilon$ and at-most-two overlap. The Lean library ships both forms (Corollaries 2.9 and 2.10 in §2.3); Theorem 1.3 is the combinatorial construction that turns local-witness data into such a cover in the 1D case.

Both results reappear in §2: Theorem 1.3 as the library's main theorem (Theorem 2.7, §2.2), and Corollary 1.6 as the one-parameter sup-reduction corollary (Corollary 2.10, §2.3).

Scope. This note formalizes the proof of the Claim (the combinatorial argument on $[0,1]$) as a Lean 4 library, `CombArg`. We do *not* formalize [DLT13, Lemma 3.1] (the replacement family), the $(U^1, U^2) \in \mathcal{CO}$ disjoint-pair structure, or the final contradiction against $m_0 = \inf_\Lambda \mathcal{F}$; these depend on the GMT and admissible-class framework of the ambient construction and remain the consumer's responsibility. The library ships one main theorem (the combinatorial core, §2.2) on $[0,1]$, together with two sup-reduction corollaries (§2.3): a generic compact-K form and its one-parameter specialization. The formalization is machine-verified against `Mathlib`, declares no custom axioms, and contains no `sorry`. Source: https://github.com/MathNetwork/comb_arg.

2. Specification

We present the library specification following standard software documentation conventions: interface first (§2.1), public API (§2.2–§2.3), internal lemmas (§2.4).

2.1. Interface: types and structures.

Throughout, `K` denotes a topological space (the parameter space) and `f : K` $\to \mathbb{R}$ a continuous scalar energy.

In the proof of Theorem 1.2, the Lemma 3.1 of [DLT13] supplies, for each near-critical parameter t, a local replacement on an $\mathcal{H}^n$-disjoint pair of open sets in M^{n+1} whose boundary energy is reduced by at least $1/(4N)$ on a neighborhood of t. The replacement family $\{\partial\Omega_{t,\tau}\}_{\tau\in[0,1]}$ at a near-critical t has its Hausdorff measure decreased by at least $1/(4N)$ inside U_t'. We keep only the scalar *replacement energy* $E : K \to \mathbb{R}$ and the uniform saving inequality; the geometric family is discarded.

Definition 2.1 (`LocalWitness` structure). [✓ `LocalWitness`]

A `LocalWitness K f t` ε carries an open neighborhood $U \ni t$ in K, a continuous replacement energy $E : K \to \mathbb{R}$, and the inequality $\forall s \in U,\ f(s) - E(s) \geq \varepsilon$.

The proof of the Claim works on the set $K \subset [0,1]$ of parameters at which the sweepout is within $1/N$ of the min-max value. We use the same convention.

Definition 2.2 (Near-critical set). [✓ `nearCritical`]

For continuous $f : K \to \mathbb{R}$, scalar m_0, and $\delta \geq 0$, the δ-near-critical set is $\{t \in K : f(t) \geq m_0 - \delta\}$. When $\delta = 1/N$ we write NC.

At the end of the refinement the data is a family $\{\overline{J_l}\}$ covering K, with each J_l carrying a replacement whose energy saving is at least $1/(4N)$ and with two-fold overlap. We abstract this output: the near-critical threshold $1/N$ becomes a parameter δ and the saving floor $1/(4N)$ a parameter $\varepsilon \leq \delta$.

Definition 2.3 (`FiniteCoverWithWitnesses` structure). [✓ `FiniteCoverWithWitnesses`]

A `FiniteCoverWithWitnesses K f m₀` δ ε carries a finite nonempty index type ι, pieces $\{p_l \subseteq K\}_{l \in \iota}$, replacement energies $\{E_l : K \to \mathbb{R}\}$, and savings $\{s_l \in \mathbb{R}_{>0}\}$, subject to four conditions: (I) $f - E_l \geq s_l$ on p_l; (II) each $t \in K$ lies in at most two pieces; (III) $s_l \geq \varepsilon$ for all l; (IV) $\{t \in K : f(t) \geq m_0 - \delta\} \subseteq \bigcup_l p_l$.

The skip-2 spacing condition (a) of the original proof is a purely one-dimensional geometric condition on a family of open intervals. We isolate it in its own structure so the disjointness lemmas that depend only on spacing can be proved independently of the witness data.

Definition 2.4 (`SkippedSpacedIntervals` structure). [✓ `SkippedSpacedIntervals`]

A `SkippedSpacedIntervals` records a natural number n, centers $c_1, \ldots, c_n \in \mathbf{I}$, positive radii $r_1, \ldots, r_n > 0$, and the skip-2 spacing axiom $c_i + r_i < c_{i+2} - r_{i+2}$ whenever $i + 2 \leq n$. The i-th open interval $I_i = (c_i - r_i, c_i + r_i) \cap \mathbf{I}$ is an abbreviation on this structure.

On top of the geometric structure, the initial cover also carries a witness center $w_i \in$ NC with an attached local witness for each interval, together with the containment of each I_i inside the corresponding witness neighborhood. In our formalization the interval center c_i and the witness center w_i are recorded separately: grid points need not themselves lie in NC.

Definition 2.5 (`InitialCover` structure). [✓ `InitialCover`]

Specialized to $K = \mathbf{I}$. An `InitialCover f m₀ N` carries the data of a `SkippedSpacedIntervals` (Definition 2.4, satisfying spacing condition (a)), plus, for each $i = 1, \ldots, n$, a witness center $w_i \in$ NC and a local witness at w_i with saving $1/(4N)$, subject to (b) the containment $I_i \subseteq U_{w_i}$ and (c) the coverage NC $\subseteq \bigcup_i I_i$.

The refinement is described as an iteration: each I_j with $j \geq 2$ is either kept whole or split into two halves. The state of this iteration after ℓ moves is the following.

Definition 2.6 (`PartialRefinement` structure). [✓ `PartialRefinement`]

Given `ic : InitialCover` and $\ell \in \mathbb{N}$, a `PartialRefinement ic` ℓ carries pieces $J_1, \ldots, J_\ell \subseteq \mathbf{I}$ and an index assignment $\sigma : \{1, \ldots, \ell\} \to \{1, \ldots, n\}$ with (i) $J_k \subseteq I_{\sigma(k)}$ for each k; (ii) $I_{\sigma(k)} \subseteq \bigcup_{k'} J_{k'}$ for each k.

2.2. **Public API: combinatorial core.** The library's main theorem is the existence of a refined multiplicity-bounded cover of the near-critical set carrying the witness savings: the non-trivial combinatorial content extracted from the original proof. Its hypotheses coincide with the intro statement (Theorem 1.3); the output is a `FiniteCoverWithWitnesses` (Definition 2.3).

Theorem 2.7 (Combinatorial core). [✓ `exists_refinement`]

Let $f : \mathbf{I} \to \mathbb{R}$ be continuous with $m_0 = \sup f$, and let $N \in \mathbb{N}$ with $N > 0$. Suppose every $t \in \mathrm{NC}$ admits a `LocalWitness` *at t with saving $1/(4N)$ (Definition 2.1). Then there exists a* `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$ *(Definition 2.3): a finite family of closed pieces $\{p_l \subseteq \mathbf{I}\}$, continuous replacement energies $\{E_l : \mathbf{I} \to \mathbb{R}\}$, and savings $\{s_l \geq 1/(4N)\}$ satisfying conditions (I)–(IV).*

The proof is given in §3 as three chained stages: Stage A builds an initial cover from a Lebesgue-number argument on NC; Stage B runs a bounded smallest-index induction to a terminal partial refinement; Stage C assembles the output by extending the open-neighborhood saving bound to the closure via continuity and deriving two-fold overlap via the parity-gap disjointness argument.

2.3. **Public API: sup-reduction corollaries.** The arithmetic that turns a `FiniteCoverWithWitnesses` into a competitor with reduced supremum is a short scalar lemma independent of covers, pieces, or multiplicity: any competitor that saves a fixed amount on the relevant super-level set yields the same amount of supremum reduction.

Lemma 2.8 (Scalar sup reduction). [✓ `csSup_range_le_of_pointwise_saving`]

Let K be a nonempty set and $f : K \to \mathbb{R}$ a bounded function with $m_0 = \sup f$. Let $g : K \to \mathbb{R}$ satisfy

(i) *$g(t) \leq f(t)$ for every $t \in K$;*
(ii) *$f(t) - g(t) \geq \varepsilon$ whenever $f(t) \geq m_0 - \delta$,*

where $0 < \varepsilon \leq \delta$. Then $\sup g \leq m_0 - \varepsilon$.

Proof. Fix $t \in K$. If $f(t) \geq m_0 - \delta$, then (ii) gives $g(t) \leq f(t) - \varepsilon \leq m_0 - \varepsilon$. If $f(t) < m_0 - \delta$, then (i) gives $g(t) \leq f(t) < m_0 - \delta \leq m_0 - \varepsilon$. Taking the supremum over t completes the proof. □

Lemma 2.8 is a general statement about lowering the supremum of a bounded real-valued function by exhibiting a competitor with a pointwise saving on the super-level set $\{f \geq m_0 - \delta\}$; no sweepout, cover, or multiplicity hypothesis enters. It applies verbatim whenever a pointwise saving can be established on the "where it matters" super-level set. The Lean library ships two corollaries of Theorem 2.7 via this lemma: the generic-K form (any `FiniteCoverWithWitnesses`), and the one-parameter specialization to $\mathbf{I}$ that chains through Theorem 2.7.

Corollary 2.9 (Sup reduction, generic form). [✓ `exists_sup_reduction_of_cover`]

Let K be a compact, nonempty topological space, $f : K \to \mathbb{R}$ continuous with $m_0 = \sup f$, and $0 < \varepsilon \leq \delta$. Given a `FiniteCoverWithWitnesses`$(K, f, m_0, \delta, \varepsilon)$ *(Definition 2.3) with pieces $\{p_l\}$, there exists $f' : K \to \mathbb{R}$ satisfying:*

(a) *$f'(t) \leq f(t)$ for every $t \in K$;*
(b) *$f'(t) = f(t)$ whenever $t \notin \bigcup_l p_l$;*
(c) *$\sup f' \leq m_0 - \varepsilon$.*

Proof. Set $f'(t) := f(t) - \sum_{l\,:\,t \in p_l} s_l$. Since each $s_l > 0$, (a) holds. If $t \notin \bigcup_l p_l$ the sum is empty, giving (b). For (c), apply Lemma 2.8: (i) follows from (a); (ii) holds because when $f(t) \geq m_0 - \delta$, (IV) gives some $p_l \ni t$, (III) gives $s_l \geq \varepsilon$, and the remaining at-most-one saving (by two-fold overlap (II)) is non-negative, so $f(t) - f'(t) \geq \varepsilon$. The lemma yields $\sup f' \leq m_0 - \varepsilon$. □

Corollary 2.10 (One-parameter sup reduction). [✓ `exists_sup_reduction`]

Under the hypotheses of Theorem 2.7, there exist $f' : \mathbf{I} \to \mathbb{R}$ and a set $S \subseteq \mathbf{I}$ containing NC *such that:*

(a) *$f'(t) \leq f(t)$ for every $t \in \mathbf{I}$;*
(b) *$f'(t) = f(t)$ whenever $t \notin S$;*
(c) *$\sup f' \leq m_0 - 1/(4N)$.*

Corollary 2.10 is Corollary 2.9 applied to the `FiniteCoverWithWitnesses` produced by Theorem 2.7, with $(\delta, \varepsilon) = (1/N, 1/(4N))$; the modification set S is the union of the cover pieces, which contains NC by condition (IV) of Definition 2.3.

2.4. **Internal lemmas.** The declarations below are the sub-lemmas of Theorem 2.7; proofs are sketched in §3.

Proposition 2.11 (Compactness of NC). [✓ `isCompact_nearCritical`, ✓ `nearCritical_nonempty`]
If K is compact and f continuous, NC *is compact. If additionally $m_0 = \sup f$ and $N > 0$,* NC *is nonempty.*

- ✓ `exists_initialCover` : existence of an `InitialCover` via a grid plus Lebesgue-number construction on the compact sublevel set NC.
- ✓ `exists_terminal_refinement` : existence of a terminal `PartialRefinement` with injective σ via bounded smallest-index induction with two-case dispatch.
- ✓ `not_three_overlap` : at most two among any three distinct `InitialCover` intervals overlap at a common point (parity rescue).
- ✓ `saving_bound_closure` : the open-neighborhood saving bound in `LocalWitness` extends to the closure via continuity of E.

The assembly itself, which packages the Stage A–B–C output as a `FiniteCoverWithWitnesses` with $(\delta, \varepsilon) = (1/N, 1/(4N))$ on $\mathbf{I}$, is Theorem 2.7 itself, whose Lean declaration ✓ `exists_refinement` chains the sub-lemmas above.

Figure 1 shows the two-tier architecture.

3. Proof structure

We split the proof of Theorem 2.7 into three stages. Stage A builds an initial cover from the witness hypothesis. Stage B runs an induction producing a terminal partial refinement with injective σ. Stage C assembles the finite cover with witnesses by extending saving bounds to closures and deriving TwoFold. The sup-reduction corollaries (Corollary 2.9 and Corollary 2.10) follow by Lemma 2.8 applied to the resulting cover; we sketch that 3-line argument after Stage C.

3.1. Stage A: existence of the initial cover.

Proposition 3.1 (Existence of an initial cover). [✓ `exists_initialCover`]
Under the hypotheses of Theorem 2.7, an initial cover (Definition 2.5) exists.

Proof. Compactness of NC (Proposition 2.11) plus the open cover $\{U_t\}_{t \in \mathrm{NC}}$ afforded by the witness hypothesis admit a Lebesgue number $\lambda > 0$: every λ-ball in NC fits inside some U_t. Choose $M \in \mathbb{N}$ with $M > 4/\lambda$, form the grid $c_k = k/M$ for $k = 0, \ldots, M$, and select those c_k within $\lambda/4$ of NC. For each selected c_k pick a witness center $w_k \in \mathrm{NC}$ within $\lambda/4$ of c_k; the ball of radius λ around w_k fits inside some U_{w_k}, hence so does the interval $I_k := (c_k - 2/(3M), c_k + 2/(3M)) \cap \mathbf{I}$. Spacing condition (a) holds because consecutive selected grid indices differ by at least 2 in the selected finset, so their centers differ by at least $2/M > 4/(3M) = 2 \cdot (2/(3M))$. Coverage condition (c) holds because every $t \in \mathrm{NC}$ is within $1/(2M) < 2/(3M)$ of its rounded grid neighbor, which is selected. Condition (b) holds by the Lebesgue argument. □

3.2. Stage B: the refinement induction.

Starting from an initial cover, one builds pieces inductively.

Definition 3.2 (Base case). [✓ `step_zero`]
step_zero(ic) is the length-1 partial refinement with $J_1 := I_1$ and $\sigma(1) := 1$.

Definition 3.3 (Inductive step). [✓ `step_succ_at`]
Given a partial refinement pr of length $\ell + 1$ and an index $i^* \in \{1, \ldots, n\}$ with $I_{i^*} \not\subseteq \bigcup_k J_k$, define a length-$(\ell + 2)$ extension by case split on whether $I_{i^*} \subseteq J_\ell \cup I_{\sigma(\ell)}$:

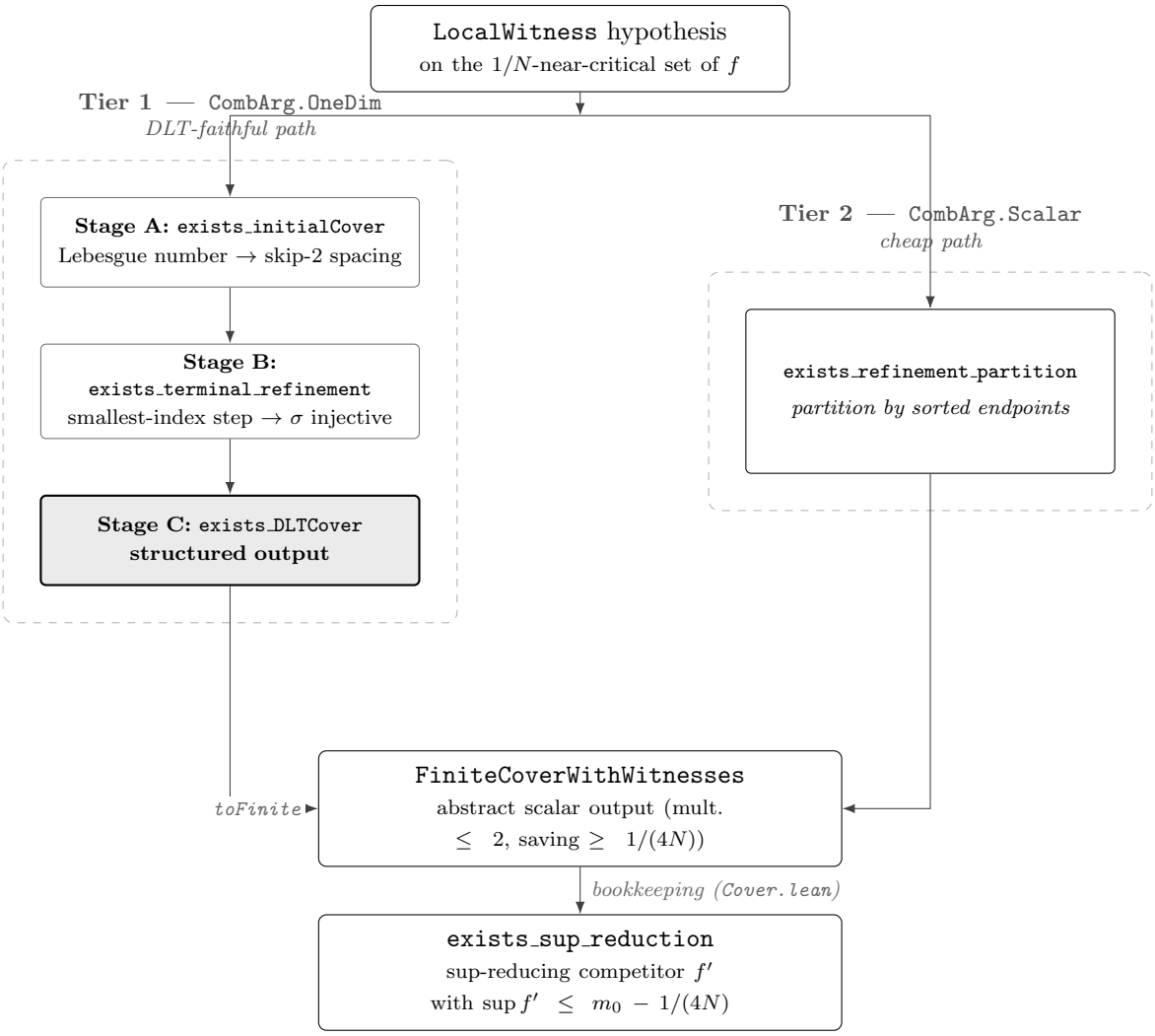

FIGURE 1. `CombArg` v0.4 two-tier architecture. The `LocalWitness` hypothesis feeds two independent proof paths producing the same abstract output. Tier 1 (`CombArg.OneDim`) follows the DLT §3.2 Step 1 algorithm in three stages culminating in the structured `DLTCover` (**bold** box), which preserves the spacing, σ-injectivity, and partial-refinement invariants needed by a future geometric lift to a modified sweepout (`CombArg/Geometric/`, not yet implemented; the lift consumes `DLTCover`, not the abstract output below). Tier 2 (`CombArg.Scalar`) proves the same conclusion directly by partition by sorted endpoints and imports none of Tier 1's spacing / refinement / parity machinery — the dependency-graph fact behind Remark 1.5. Both deliver the same `FiniteCoverWithWitnesses`, which the K-generic bookkeeping corollary in `Cover.lean` turns into the sup-reducing competitor f'.

(Case 1) If yes, set $J_{\ell+1} := I_{i^*}$.
(Case 2) Otherwise, set $J_{\ell+1} := I_{i^*} \setminus I_{\sigma(\ell)}$.

In both cases set $\sigma(\ell+1) := i^*$. Both invariants of Definition 2.6 are preserved; in Case 2 the preservation of (ii) uses the induction hypothesis on $I_{\sigma(\ell)} \subseteq \bigcup_k J_k$.[2]

[2]A variant `step_succ` chooses i^* internally via Finset.min′; `step_succ_at` takes i^* as an explicit argument and returns the extension bundled with the "i^* is covered" property, for termination.

Proposition 3.4 (Termination). [✓ `exists_terminal_refinement`]

Starting from step_zero(ic) *and iterating* step_succ_at, *one reaches in at most n steps a partial refinement of some length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i.*

Proof. Let rem(pr) be the finite set of indices i with $I_i \not\subseteq \bigcup_k J_k$. One step of step_succ_at at $i^* \in$ rem(pr) (chosen as the smallest such) produces pr′ with $i^* \notin$ rem(pr′) and rem(pr′) $\subseteq$ rem(pr)$\setminus\{i^*\}$, hence strict decrease. Injectivity of σ follows from the case split in Definition 3.3: the new $\sigma(\ell+1) = i^*$ is fresh because $i^* \in$ rem while existing $\sigma(k)$ correspond to already-covered indices. □

3.3. **Stage C: assembly.** Stage C packages the refined covering as a `FiniteCoverWithWitnesses` (Definition 2.3), which the core theorem turns into the sup-reducing f'. We trace through [DLT13, §3.2, Step 2] in sequence, indicating at each step where the content lands in the formalization.

De Lellis and Tasnady begin by applying Lemma 3.1 to produce, at each near-critical parameter t, a replacement family $\{\partial\Omega_{i,t}\}$ with three properties: $\Omega_{i,t} = \Omega_t$ outside (a_i, b_i) and agrees with Ω_t outside U_i' inside (a_i, b_i); the global upper bound $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) + \frac{1}{4N}$; and, most importantly, the saving $\mathcal{H}^n(\partial\Omega_{i,t}) \le \mathcal{H}^n(\partial\Omega_t) - \frac{1}{2N}$ when $t \in (a_i + \delta, b_i - \delta)$. In our formalization this data is what a `LocalWitness` (Definition 2.1) carries: the scalar $E_{\sigma(k)} : K \to \mathbb{R}$ plays the role of $\mathcal{H}^n(\partial\Omega_{i,t})$, the neighborhood U_t plays (a_i, b_i), and the two separate bounds ("saving $-\frac{1}{2N}$ on the inner core" and "no-loss $+\frac{1}{4N}$ outside") collapse to a single uniform saving $\varepsilon = \frac{1}{4N}$ on U_t.[3] Stage B has already handed us pieces p_l on which this witness saving is valid (Proposition 3.6), so by Stage C we have per-piece scalar savings $s_l \ge \varepsilon$ directly.

The next observation in the proof is that if $t \in (a_i, b_i) \cap (a_j, b_j)$, then $j = i + 1$, and moreover $\operatorname{dist}(U_i', U_{i+1}') > 0$. This is the two-fold overlap of the refined covering, and in our formalization it is the `twoFold` field of `FiniteCoverWithWitnesses`, derived from the parity-gap spacing by Proposition 3.5. The parity-gap lemmas themselves live on the abstract `SkippedSpacedIntervals` (Definition 2.4) and are therefore independent of the witness data: `InitialCover` provides access to them by projecting out the underlying geometry.

The modified sweepout $\{\partial\Omega_t'\}$ is then defined piecewise: $\Omega_t' = \Omega_t$ outside all J_i; $\Omega_t' = \Omega_{i,t}$ if t lies in a single J_i; and on $J_i \cap J_{i+1}$ the sweepout blends the two replacements, $\Omega_t' = [\Omega_t \setminus (U_i' \cup U_{i+1}')] \cup [\Omega_{i,t} \cap U_i'] \cup [\Omega_{i+1,t} \cap U_{i+1}']$. Stripped of the geometry this is additive scalar subtraction of savings over the pieces containing t: our reduced energy is

$$f'(t) \;=\; f(t) \;-\; \sum_{l\,:\,t\in\overline{J_l}} s_l,$$

and the two-fold invariant bounds the sum length by 2, matching the two-piece blend on overlaps.

The area accounting then splits on whether t is near-critical. For $t \notin K$, the point lies in at most two J_i's and each replacement can add at most $\frac{1}{4N}$, so

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) + \tfrac{1}{2N}.$$

For $t \in K$, the point lies in at least one inner core $(a_i + \delta, b_i - \delta)$, where the replacement saves $\ge \frac{1}{2N}$ in U_i', and in at most one other J_l where it costs $\le \frac{1}{4N}$ in U_l', for net

$$\mathcal{H}^n(\partial\Omega_t') \;\le\; \mathcal{H}^n(\partial\Omega_t) - \tfrac{1}{4N}.$$

In our formalization the same case analysis is the scalar argument for $f'(t) \le m_0 - \varepsilon$ inside the abstract core (the lemma `reducedEnergy_le` from §2.2): split on whether t lies in any piece of the cover, use the saving floor $s_l \ge \varepsilon$ in the positive branch and the covering condition $\{f \ge m_0 - \delta\} \subseteq$

[3] This is strictly stronger than the original two-bound statement: the local witness asserts a uniform saving of $\frac{1}{4N}$ on all of U_t, whereas De Lellis–Tasnady only require a larger saving $\frac{1}{2N}$ on the inner core $(a_i + \delta, b_i - \delta)$ and bounded loss $\frac{1}{4N}$ elsewhere. The stronger abstract assumption removes the δ parameter of the inner core from the scalar theorem; the caller must verify the uniform version holds in the ambient setting.

$\bigcup_l p_l$ in the negative branch. With $(\delta, \varepsilon) = (1/N, 1/(4N))$ this specializes to the arithmetic of the original proof; the abstract core reads off the same algebra for arbitrary (δ, ε) with $\varepsilon \le \delta$.

The rest of this subsection establishes the two propositions that Stage B leaves open: the saving bound extends from open J_k to $\overline{J_k}$ (continuity), and the two-fold invariant holds (a parity argument on the spacing condition (a)).

Proposition 3.5 (TwoFold via parity). [✓ `terminal_twoFold`]

For a partial refinement pr *of length L with σ injective, every $t \in \mathbf{I}$ lies in at most two of $\overline{J_1}, \dots, \overline{J_L}$.*

Proof. From spacing condition (a), iterated m times, one obtains that indices i, j with $|i-j| = 2(m+1)$ (an even gap ≥ 2) have strict separation between $\overline{I_i}$ and $\overline{I_j}$, hence $\overline{I_i} \cap \overline{I_j} = \emptyset$ (✓ `chain_spacing`, ✓ `closure_disjoint_of_even_gap`).

Suppose three distinct indices $k_1, k_2, k_3 \in \{1, \dots, L\}$ admit $t \in \overline{J_{k_1}} \cap \overline{J_{k_2}} \cap \overline{J_{k_3}}$. Since $\overline{J_k} \subseteq \overline{I_{\sigma(k)}}$ and σ is injective, $\sigma(k_1), \sigma(k_2), \sigma(k_3)$ are three distinct indices in $\{1, \dots, n\}$ with a common point in the closures of $I_{\sigma(k_i)}$. Among any three distinct integers $a < b < c$ some pair has even gap ≥ 2: either $c - a$ is even (and necessarily ≥ 2), or $c - a$ is odd and ≥ 3, in which case $b - a$ and $c - b$ have opposite parities and one of them is the even gap. The pair with even gap has disjoint closures, contradicting the common point (✓ `not_three_overlap`). □

Proposition 3.6 (Closure extension of saving bound). [✓ `saving_bound_closure`]

Let pr *be a partial refinement. For every k and every $t \in \overline{J_k}$, $f(t) - E_{\sigma(k)}(t) \ge 1/(4N)$, where $E_{\sigma(k)}$ is the replacement energy of the $\sigma(k)$-th local witness.*

Proof. On $J_k \subseteq U_{w_{\sigma(k)}}$ the witness gives $f - E_{\sigma(k)} \ge 1/(4N)$. The function $g := f - E_{\sigma(k)}$ is continuous (Definitions 2.1 requires E_t continuous; f is continuous by hypothesis). A closure point is a limit of a sequence in J_k, and g preserves $\ge$-inequalities under limits. □

Proof of Theorem 2.7. Chain Propositions 3.1 and 3.4: the former produces an `InitialCover` from the witness hypothesis, and the latter yields a terminal `PartialRefinement` pr of length L with σ injective and $I_i \subseteq \bigcup_k J_k$ for every i. Assemble the `FiniteCoverWithWitnesses`$(\mathbf{I}, f, m_0, 1/N, 1/(4N))$ by taking pieces $p_k := \overline{J_k}$, uniform saving $s_k := 1/(4N)$, and E_k the replacement energy of the $\sigma(k)$-th local witness. Invariant (I) is Proposition 3.6; invariant (II) is Proposition 3.5; invariant (III) holds by construction ($s_k = 1/(4N)$); coverage of NC chains as $\mathrm{NC} \subseteq \bigcup_i I_i \subseteq \bigcup_k J_k \subseteq \bigcup_k \overline{J_k}$. □

Corollary 2.9 then follows by the three-line bookkeeping proved in §2.3 applied to this `FiniteCoverWithWitnesses`; Corollary 2.10 is the specialization to $(\delta, \varepsilon) = (1/N, 1/(4N))$.

4. Lifting to a min-max proof

We indicate how the formalization developed in §§2–3 fits into a complete formalization of an Almgren–Pitts min-max construction. Suppose the surrounding geometric and admissibility infrastructure has been set up in Lean: an admissible class $\mathcal{A}$ of sweepouts, the inf-sup characterization $m_0 = \inf_{\Omega \in \mathcal{A}} \sup f_\Omega$, the local replacement construction [DLT13, Lemma 3.1], and the Hausdorff measure of boundary surfaces. The four objects introduced in §2, `LocalWitness` (Definition 2.1), ✓ `exists_refinement`, ✓ `exists_sup_reduction_of_cover`, and the chained ✓ `exists_sup_reduction`, carry no geometric content. They are, respectively, the abstract local-reducer data, [DLT13] §3.2 Step 1 (the interval refinement to a finite cover with two-fold overlap), [DLT13] §3.2 Step 2 (the scalar bookkeeping over the cover), and the chained 1-parameter conclusion. The same four declarations apply in both the abstract and the instantiated form; what changes between them is only the data passed in.

4.1. **The geometric input: `LocalWitness` from the local replacement lemma.** The `LocalWitness` structure was chosen so that its fields match the data the local replacement lemma [DLT13, Lemma 3.1] produces. Fix a sweepout $\Omega \in \mathcal{A}$ realizing $m_0 = \sup f_\Omega$ and an integer $N \geq 1$. At every parameter $t \in [0,1]$ with $f_\Omega(t) \geq m_0 - 1/N$, the replacement lemma supplies an open interval $(a_t, b_t) \subseteq [0,1]$ around t on which a local replacement saves boundary energy. Recording the interval as the `neighborhood` field, the boundary energy of the replaced sweepout

$$s \longmapsto \mathcal{H}^n(\partial\widetilde{\Omega}_s), \qquad \widetilde{\Omega}_s := \Omega \text{ with the local replacement inserted at parameter } s,$$

as the `replacementEnergy` field, its continuity in s as `replacementEnergy_continuous`, and the quantitative estimate

$$f_\Omega(s) - \mathcal{H}^n(\partial\widetilde{\Omega}_s) \;\geq\; \frac{1}{4N} \qquad \text{for } s \in (a_t, b_t)$$

as `saving_bound`, one obtains an instance of `LocalWitness` $([0,1], f_\Omega, t, 1/(4N))$. Of these four data items, only the continuity is treated implicitly in the literature (through the continuity of the replaced family in the inserted parameter); in the Lean formalization it must be supplied explicitly.

4.2. **The chain: from sup reduction to inf-sup contradiction.** With the per-parameter local witnesses in place, the chained theorem ✓ `exists_sup_reduction` produces a competitor $f' : [0,1] \to \mathbb{R}$ together with a modification set $S \subseteq [0,1]$ satisfying

$$\{t : f_\Omega(t) \geq m_0 - 1/N\} \subseteq S, \qquad f' \leq f_\Omega \text{ on } [0,1], \qquad f' = f_\Omega \text{ on } [0,1] \setminus S,$$

together with the supremum bound $\sup f' \leq m_0 - 1/(4N)$. This is the formalization of the sup-reduction step in the literature argument: f' is the boundary energy of Ω after the local replacements have been combined, and the off-S agreement ensures that f' is a genuine combinatorial reduction of f_Ω rather than a trivial constant.

The proof of the min-max theorem now closes the standard way. The off-S agreement makes the geometric lift routine: there exists $\Omega' \in \mathcal{A}$ that agrees with Ω outside S and uses the local replacement on each component of S, and one checks $f' = f_{\Omega'}$. The three inequalities

$$\sup f_{\Omega'} \;=\; \sup f' \;\leq\; m_0 - \tfrac{1}{4N} \;<\; m_0 \;=\; \inf_{\mathcal{A}} \sup$$

together with $\Omega' \in \mathcal{A}$ contradict the inf-sup characterization of m_0. The first inequality is the present formalization; the geometric lift producing Ω' is supplied by the surrounding Lean formalization. The full Lean script for this chain appears as Listing 1 in Appendix A.

4.3. **Scope of the present formalization.** We close with three caveats. First, the parameter space is fixed at $[0,1]$; multi-parameter sweepouts ($[0,1]^m$, in particular Almgren cycles) require a multi-parameter cover construction not carried out here. Second, the output is the scalar pair (f', S), not a geometric replacement: the construction of the sweepout $\Omega' \in \mathcal{A}$ realizing $f' = f_{\Omega'}$ falls outside the present formalization, though the off-S agreement was chosen so as to make this lift mechanical. Third, the continuity of the replaced family in the inserted parameter is treated implicitly in [DLT13]; in the Lean formalization this becomes the `replacementEnergy_continuous` field of `LocalWitness` and must be supplied explicitly. The formalization itself depends only on the three standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`; standard measure-theory work in Mathlib stays within the same three.

5. Generality across one-parameter min-max settings

The combinatorial step formalized here is not specific to the closed-minimal-hypersurface setting of [DLT13]. Recall the shape of the 1-parameter Almgren–Pitts pattern: a continuous family of sweepouts indexed by $t \in [0,1]$, a scalar energy $f(t)$ attached to each sweepout, the inf-sup level $m_0 = \inf_{\mathcal{A}} \sup f$ over an admissible class $\mathcal{A}$, a local replacement lemma producing a quantitative saving on a neighborhood of each near-critical t, and a combinatorial step combining these local

replacements into a competitor strictly below m_0. This pattern recurs in several geometric variants beyond the original. We mention four.

- *Free boundary minimal hypersurfaces* [LZ21]: the sweepout is by free-boundary regions in a manifold with boundary, and the local replacement lemma uses level sets of the distance function to the hypersurface in place of the exponential-map construction of [DLT13]. The energy is again the boundary $\mathcal{H}^n$-area.
- *Constant mean curvature hypersurfaces* [ZZ19]: the energy is the prescribed-mean-curvature functional $f_\Omega = \mathcal{H}^n(\partial\Omega) - c \cdot \mathrm{vol}(\Omega)$, and the local replacement lemma is adapted to produce a quantitative saving in this functional.
- *Complete manifolds of finite volume* [CL20]: the sweepout is by hypersurfaces of bounded volume in a complete non-compact manifold; the main geometric input is the disjointing lemma stating that any such sweepout can be made by mutually disjoint hypersurfaces of nearly the same volume.
- *Anisotropic min-max* [DPDRL24]: the area $\mathcal{H}^n(\partial\Omega)$ is replaced by an elliptic integrand $F(\partial\Omega)$; the local replacement lemma carries the anisotropic structure and produces an analogous quantitative saving.

In each case the local replacement lemma differs in substance from the one in [DLT13], but the subsequent combinatorial step is the same: the interval refinement, the two-fold overlap invariant, and the scalar bookkeeping that turns the resulting cover into a sup-reducing competitor are all carried out by the formalization presented here. The `LocalWitness` structure of §4.1 does not depend on the geometric origin of its four data fields, specifying only an open neighborhood, a continuous scalar energy, and the quantitative saving inequality. Any of the local replacement lemmas above can therefore feed the present formalization to produce a sup-reducing competitor for the corresponding inf-sup level, with the script of Appendix A carrying over verbatim and only the geometric stubs (`YourGMT.*`) replaced by the relevant variant's geometric formalization.

6. Future work

Two natural extensions sit immediately downstream of the present formalization, both prepared for by the v0.4 two-tier architecture of §3.

6.1. **Geometric lift to a modified sweepout.** The structured output of §3 (✓ `exists_DLTCover`) packages the Stage A initial cover with spacing condition (a), the Stage B partial refinement, σ-injectivity, and the termination invariant $\forall i,\ I_i \subseteq \bigcup_k J_k$ as a single Lean structure `DLTCover`. The companion lift

$$\texttt{DLTCover} \longrightarrow \{\partial\Omega'_t\}_{t\in[0,1]} \text{ flat-continuous}$$

realizes the De Lellis–Tasnady modified sweepout Ω'_t defined by cut-paste blending on the positive-measure overlaps $J_i \cap J_{i+1}$. The blending formula

$$\Omega'_t \;=\; [\Omega_t \setminus (U'_i \cup U'_{i+1})] \;\cup\; [\Omega_{i,t} \cap U'_i] \;\cup\; [\Omega_{i+1,t} \cap U'_{i+1}]$$

is purely set-algebraic, and its t-continuity in the flat metric follows from t-continuity of the input families $\{\Omega_t\}, \{\Omega_{i,t}\}$ together with t-independence of the cut sets U'_i, U'_{i+1}; no smooth structure or partition of unity is required. The lift therefore becomes formally available once an ambient Lean library supplies a finite-perimeter / sweepout API with a flat-continuity notion — a strictly geometric prerequisite, separate from the combinatorial argument formalized here.

The placeholder directory `CombArg/Geometric/` marks the import position the future lift module will occupy. Remark 1.5 is the design contract: `DLTCover`'s preserved spacing/overlap structure, not the abstract scalar bound of Corollary 2.10, is what the geometric lift consumes; the cheap partition route (`CombArg.Scalar.exists_refinement_partition`) shipped as its companion verifies by dependency-graph fact that the DLT structure is essential exactly at this lift step and at no earlier one.

6.2. **Multi-parameter sweepouts.** The cover construction in §3 is specialized to the 1-parameter case $K = [0,1]$. Multi-parameter min-max constructions ($K = [0,1]^m$ for $m \geq 2$, in particular Almgren cycles in Marques–Neves [MN14]) replace the skip-2 spacing of §3 (an inequality on 1-dimensional indices) with a corresponding spacing structure on an m-dimensional grid; the parity rescue (§3, Proposition 3.5) that delivers the two-fold overlap invariant from skip-2 spacing needs an m-dimensional analogue. The bookkeeping corollary ✓ `exists_sup_reduction_of_cover` is already K-generic in compact K, so only the cover construction itself must be redone for the multi-parameter setting.

The natural development is a sister namespace `CombArg.MultiDim` parallel to the present `CombArg.OneDim`, producing a `DLTCover`-analogous structured output for each $m \geq 1$ that the same bookkeeping corollary then turns into a sup-reducing competitor on $K = [0,1]^m$. The two-tier architecture inherited from the 1-dimensional case — a structured DLT-style cover preserving geometric data and a cheaper alternative scalar proof for the same abstract conclusion — is expected to carry over: in each dimension, the abstract scalar reduction admits a partition-style shortcut, while the geometric lift to a multi-parameter modified sweepout consumes the structured form.

Together, the two directions complete the formalization arc sketched at the start of §4: this paper formalizes the combinatorial slice; §6.1 crosses into geometric measure theory once an ambient library is in place; and §6.2 extends the combinatorial slice itself across parameter dimensions. Each is independently scoped and does not block the other.

Appendix A. End-to-end Lean script

The Lean script in Listing 1 below realizes the chain of §4.2. Identifiers prefixed by `YourGMT.*` refer to the surrounding formalization's geometric machinery: the sweepout type, the admissible class, the choice of the near-criticality parameter N, the local replacement lemma, and the geometric lift back to a sweepout. The `CombArg.*` identifier is the chained sup-reduction theorem of the present paper.

The script reads from top to bottom as the three steps of §4.2. The per-parameter local witness is first constructed from the geometric replacement lemma. The combinatorial step of this paper is then invoked to produce the sup-reducing competitor (f', S). Finally, the geometric lift to $\Omega' \in \mathcal{A}$, followed by the inf-sup contradiction, occupies the closing block.

Listing 1. End-to-end script for a min-max contradiction.

```
import CombArg
import YourGMT

theorem minmax_contradiction
    (O : YourGMT.SweepoutType) (hO : O in YourGMT.admissibleClass)
    (h_minmax : O.boundaryEnergy.sup = YourGMT.infMinmax) :
    False := by
  set f := O.boundaryEnergy
  set m0 := sSup (Set.range f)
  obtain <N, hN_pos, hN_window> := YourGMT.choose_N O
  -- Per-parameter local witness from the local replacement lemma.
  have witness : forall t : unitInterval, f t >= m0 - 1 / (N : R) ->
                   LocalWitness unitInterval f t (1 / (4 * (N : R))) :=
    fun t ht => YourGMT.localWitness_of_DLT O t ht hN_pos
  -- Combinatorial step (this paper) -> scalar sup reduction.
  obtain <f', S, _, h_le, h_eq, h_sup> :=
    CombArg.exists_sup_reduction
      O.boundaryEnergyContinuous rfl hN_pos witness
```

```
-- Geometric lift back to a sweepout in admissibleClass.
obtain <O', hO', hO'_energy> :=
  YourGMT.lift_sweepout f' h_le h_eq O hO
-- Contradiction with the inf-sup characterization of m0.
have h_lt : O'.boundaryEnergy.sup < m0 := by
  rw [hO'_energy]; linarith [h_sup, hN_window]
exact absurd h_lt (YourGMT.le_of_admissible hO' h_minmax)
```

References

[CDL03] T. H. Colding and C. De Lellis. The min-max construction of minimal surfaces. *Surv. Differ. Geom.*, VIII:75–107, 2003.

[CL20] G. R. Chambers and Y. Liokumovich. Existence of minimal hypersurfaces in complete manifolds of finite volume. *Invent. Math.*, 219(1):179–217, 2020. arXiv:1609.04058.

[DL16] C. De Lellis. Exposition of the combinatorial step in the min-max construction, 2016. Annales de l'Institut Fourier, 2016. Precise article reference to be confirmed against the author's library.

[DLR18] C. De Lellis and J. Ramic. Min-max theory for minimal hypersurfaces with boundary. *Ann. Inst. Fourier*, 68(5):1909–1986, 2018.

[DLT13] C. De Lellis and D. Tasnady. The existence of embedded minimal hypersurfaces. *J. Differential Geom.*, 95(3):355–388, 2013.

[DPDRL24] G. De Philippis, A. De Rosa, and Y. Li. Existence and regularity of min-max anisotropic minimal hypersurfaces. Preprint, arXiv:2409.15232, 2024.

[LZ21] M. M.-c. Li and X. Zhou. Min-max theory for free boundary minimal hypersurfaces, I: regularity theory. *J. Differential Geom.*, 118(3):487–553, 2021.

[MN14] F. C. Marques and A. Neves. Min-max theory and the willmore conjecture. *Ann. of Math. (2)*, 179(2):683–782, 2014.

[Pit81] J. T. Pitts. *Existence and regularity of minimal surfaces on Riemannian manifolds*, volume 27 of *Mathematical Notes*. Princeton University Press, 1981.

[SS81] R. Schoen and L. Simon. Regularity of stable minimal hypersurfaces. *Comm. Pure Appl. Math.*, 34(6):741–797, 1981.

[ZZ19] X. Zhou and J. J. Zhu. Min-max theory for constant mean curvature hypersurfaces. *Invent. Math.*, 218(2):441–490, 2019.